Unit -1 Introduction to Cyber Security

Cyberspace is the virtual world where people connect, share, work, and communicate through the internet. It includes everything we interact with online—websites, apps, social media, emails, digital banking, and games. This space does not have physical borders, but it impacts our lives as strongly as the real world does. From online shopping to international video calls, modern human interaction thrives in cyberspace. As technology keeps advancing, cyberspace is expanding too. It connects billions of users daily and has become the foundation for education, business, entertainment, and social life. It supports how people access information, make money, and build digital communities. Cyberspace is central to human progress, blending digital tools with real-world needs.

What is Cyberspace?

Cyberspace is the online environment of computers, servers, networks, and digital platforms that allow people to interact without being physically present. It's where emails are sent, messages are posted on social media, and files are stored in cloud services. Every time people use a mobile app or visit a website, they operate in cyberspace. This digital domain powers everything from video calls and online exams to business meetings and bank transfers.

Although it is not something you can touch, cyberspace is real and powerful. It allows governments to operate more efficiently, businesses to run without physical stores, and students to learn from home. It has broken the boundaries of geography and time, letting people do more, faster, and with fewer limitations. It is the most significant change to human interaction since the invention of electricity.

Defining Cyberspace
What is Cyberspace?

- Cyberspace refers to the **virtual digital environment** created by interconnected computers, networks, devices, and data.
- It is not a physical space, but a **digital world** where:
    - Communication happens
    - Data is stored and shared
    - Services and applications operate

Key Characteristics:

- **Borderless** – No geographical boundaries.
- **Interactive** – Users communicate and share data in real-time.
- **Global** – Devices across the world can connect.
- **Virtual** – Exists through the internet infrastructure, not physically.

Examples of cyberspace:

- Social media platforms (Instagram, Facebook)
- Email systems
- Online banking portals
- Cloud services
- E-commerce sites (Amazon, Flipkart)

Why is Cyberspace Important?

- Enables online learning, communication, business, banking, entertainment.
- But also brings risks like cybercrimes, hacking, and privacy issues.

What is Cyberspace? (Quick Reminder)

Cyberspace = **The virtual digital world created by the Internet**, where:

- People interact
- Data is stored
- Applications run
- Transactions happen

It has **no physical shape**, but it exists through computers, networks, servers, and applications.

Example: Using Social Media (Instagram / Facebook / Twitter)

Why is it part of cyberspace?

Because you interact with **digital profiles, digital content, and digital networks**—not in physical space.

Detailed Explanation:

1. You open Instagram on your mobile.
2. You scroll through posts, reels, and stories.
3. You like, comment, chat, or follow someone.
4. Everything you see (images, videos, messages) is stored on Instagram's servers in data centres.
5. You don't meet the person physically—your interaction happens **in the virtual environment**, i.e., cyberspace.

What is happening behind the scenes?

- Your mobile connects to the internet.
- Instagram servers send data to your phone.
- Your actions (like/comment) go back to Instagram servers.
- Other users see your actions from different countries.

This is a perfect example of cyberspace because the entire "interaction" is virtual.

Example: Online Banking / UPI (Google Pay, PhonePe)
Why is it cyberspace?
Money transfer happens digitally—not through physical cash movement.
Detailed Explanation:

1. You open Google Pay.
2. You enter an amount and send it to someone.
3. Your bank verifies your identity online.
4. The transaction is processed through secure internet channels.
5. The receiver gets the amount instantly.

Behind the scenes:

- Encrypted financial data travels through the internet.
- Bank servers validate and update balances.
- No physical money or paper is involved.

All this happens in a **virtual financial environment**, which is part of cyberspace.

Example: Online Shopping (Amazon, Flipkart)
Why is it cyberspace?
You browse and buy products in a digital marketplace.
Detailed Explanation:

1. You open Amazon.
2. You search for a mobile phone.
3. Server shows product details, reviews, price comparison.
4. You add items to cart and pay online.
5. Data flows between your device, Amazon servers, and payment gateways.

Behind the scenes:

- The website is hosted on cloud servers.
- Every click sends a request to Amazon's data center.
- Your order and address are stored digitally.

The entire shopping activity (before delivery) occurs **inside cyberspace**.

Example: Email Communication (Gmail / Outlook)
Why is it cyberspace?
Sending a mail = sending digital packets over the internet.
Detailed Explanation:

1. You type an email to your friend.
2. Gmail converts the text into digital data.
3. The data is sent to Google's servers.
4. Your friend receives the email on another device.

Behind the scenes:

- SMTP protocol sends the email.
- Data is stored on cloud servers.
- Your email exists only in cyberspace (data form).

Email is a pure cyberspace interaction—no paper, no physical activity.

Example: Watching YouTube / Netflix
Why is it cyberspace?
You consume digital content stored on servers across the world.
Detailed Explanation:

1. You click a video.
2. YouTube sends video data in packets to your device.
3. Your device buffers and plays the video.
4. The video may be stored thousands of kilometers away.

Behind the scenes:

- CDN servers deliver video quickly.
- Streaming protocols manage quality.
- No physical medium is used.

The entire entertainment environment is inside cyberspace.

Example: Cloud Storage (Google Drive, OneDrive)
Why is it cyberspace?
Your files are stored digitally on remote servers—not on your local device.
Detailed Explanation:

1. You upload a PDF to Google Drive.
2. The file travels through the internet to Google's global servers.
3. You can access it from any device later.
4. You can share it with others instantly.

Behind the scenes:

- The file is broken into data chunks.
- Stored in multiple locations for safety.
- Access control decides who can view/edit.

Cloud services represent the **core of cyberspace**—virtual storage replacing physical storage.

Example: Online Gaming / Virtual Worlds (PUBG, Free Fire, Fortnite)
Why is it cyberspace?
Gamers interact inside a digital virtual world.
Detailed Explanation:

1. You play PUBG with someone in another country.
2. You both are in the same digital environment (map, team, missions).
3. You communicate through real-time voice/chat.
4. The game servers track your actions.

Behind the scenes:

- Game servers manage thousands of players.
- Network latency affects gameplay.
- Game state is updated every second.

A virtual world shared by many users = cyberspace.

Overview of Computer and Web Technology
A. Computer Technology
Components of a Computer System:

1. **Hardware:**
    - Input devices → Keyboard, mouse
    - Output devices → Monitor, printer
    - Storage → HDD, SSD

- Processor → CPU (brain of the computer)

2. **Software:**
   - **System Software** → Windows, Linux
   - **Application Software** → MS Office, Browsers, Games
3. **Networks:**
   - LAN, WAN, Wi-Fi, Ethernet
4. **Data Storage and Processing:**
   - Cloud storage, Databases, File systems

B. Web Technology

Web technologies enable websites, online services, and applications to function.

Key Components:

1. **Web Browsers:**
   - Chrome, Firefox, Edge
   - Used to access websites
2. **Web Servers:**
   - Apache, Nginx, IIS
   - Store and deliver web pages to users
3. **Web Pages & Languages:**
   - HTML → Structure
   - CSS → Design
   - JavaScript → Interaction
   - PHP / Python / Node.js → Backend
4. **Protocols:**
   - **HTTP/HTTPS** → Website communication
   - **FTP** → File transfer
   - **SMTP** → Email sending

How the Web Works (Simple Explanation):

1. User types a URL in the browser
2. Browser sends request to server
3. Server finds the page and sends it back
4. Browser displays the content to user

This entire process happens in milliseconds.

Architecture of Cyberspace and Its Role in Communication Systems

Architecture of Cyberspace

Cyberspace refers to the virtual environment of information and communication technologies where digital data is created, shared, and exchanged. The architecture of cyberspace can be understood as the layered structure that supports digital communication and interaction. It typically includes the following layers:

1. **Physical Layer**

   The Physical Layer is the most basic layer of cyberspace. It includes hardware devices such as computers, routers, servers, data centers, optical fiber cables, satellites, and communication towers. These devices form the infrastructure that transmits digital signals.
   - Routers and switches route data packets.
   - Servers store and manage data.
   - Optical fiber cables are used for high-speed data transmission.

2. **Network Layer**
   - Includes protocols like IP (Internet Protocol) that enable devices to communicate across networks.
   - The Network Layer controls data communication. It includes the Internet Protocol (IP), TCP, routers, and network addressing systems. When a user sends data online, this layer divides the data into small packets and delivers them to the destination.

    - This layer is the most crucial for data transfer because it maintains reliability and efficiency.
3. **Application Layer**
    - Provides services and applications that users interact with, such as web browsers, email clients, social media platforms, and cloud services.
    - Protocols like HTTP, FTP, SMTP operate at this layer.
    - The Application Layer is the part where users directly interact. Websites, mobile applications, email systems, and social media platforms are part of this layer.
    - Web browsers (such as Chrome, Edge, and Firefox)
    - Applications (such as Facebook, Instagram, and Gmail)
    - Protocols (such as HTTP, HTTPS, and SMTP)
4. **Content Layer**
    - Contains the actual data, information, and digital content shared and accessed in cyberspace.
    - The Data Layer is the heart of cyberspace. It includes the entire system of data creation, storage, access, and retrieval. Databases, cloud storage, and data warehouses belong to this layer.
    - This layer ensures that data remains accurate, secure, and accessible.
5. **User Layer**
    - Represents the human users who create, consume, and interact with digital content.
    - The User Layer is the most dynamic part of cyberspace. Users, developers, network administrators, and cyber experts are the real operators of this digital world.
    - The User Layer makes cyberspace meaningful because without users, no digital architecture can function.

Role of Cyberspace in Communication Systems

- **Global Connectivity:** Cyberspace enables devices and users worldwide to connect and communicate seamlessly, overcoming geographical barriers.
- **Information Exchange:** It facilitates rapid and efficient exchange of information through emails, instant messaging, video conferencing, and social media.
- **Data Sharing and Collaboration:** Supports collaborative work environments where multiple users can share and edit documents and projects in real-time.
- **Service Delivery:** Enables delivery of various communication services such as VoIP, streaming, cloud computing, and online gaming.
- **Security and Privacy:** Cyberspace architecture incorporates security protocols and encryption to protect data integrity and user privacy.
- **Innovation Platform:** Acts as a foundation for developing new communication technologies and applications.

The architecture of cyberspace provides a structured framework that supports the complex interactions and data flows necessary for modern communication systems, making global digital communication possible and efficient.

1. Physical Layer — the foundation
What it is: the tangible hardware that makes cyberspace possible.
Components / protocols: cables (copper, fiber), switches, routers, modems, Wi-Fi access points, cell towers, data center racks, batteries/UPS.
What happens: electrical/optical signals physically carry bits between devices and between data centers. Submarine cables cross oceans; local ISPs connect homes to the global network.
Security risks: physical tampering, cable cuts, theft, power outages, insider risks at data centers.
Demo idea: show a router and an Ethernet cable; trace how a packet leaves a laptop physically to the ISP.

2. Link & Network Infrastructure Layer — addressing and routing

What it is: how devices are addressed and how packets find their way across networks.
Components / protocols: MAC addresses, Ethernet/Wi-Fi link protocols, IP (IPv4/IPv6), routers, DNS, BGP (inter-ISP routing).
What happens: devices have local addresses; IP assigns global addresses; routers forward packets based on IP; DNS translates human names (example.com) to IPs. BGP decides which inter-ISP routes to use.
Security risks: IP spoofing, DNS hijacking, route hijacks (BGP), man-in-the-middle at the network layer.
Demo idea: show ping and traceroute from a classroom computer to illustrate hops and routing.

3. Transport Layer — reliable delivery between endpoints
What it is: protocols that manage communication between two endpoints with reliability, ordering, and flow control.
Components / protocols: TCP (reliable), UDP (low-latency), TLS (encryption layer used with TCP).
What happens: TCP does connection setup, acknowledgements, retransmission; UDP sends packets without guarantees (used for video/voice/gaming). TLS adds encryption on top to protect confidentiality and integrity.
Security risks: TCP/IP attacks (SYN flood/DDoS), interception if TLS not used, UDP amplification.
Demo idea: show the difference by transferring a small file via a TCP-based tool and streaming audio over UDP.

4. Service / Application Layer — the visible services
What it is: websites, APIs, email, messaging and other apps users interact with.
Components / protocols: HTTP/HTTPS, SMTP (email), WebSocket (real-time), REST APIs, GraphQL, DNS records (service points). Web servers (Apache/Nginx), application servers, microservices.
What happens: browsers request web pages over HTTP(S); servers run application code, query databases, and return responses (HTML/JSON). APIs let services talk to each other.
Security risks: XSS, SQL injection, insecure APIs, broken authentication, misconfigured servers.
Demo idea: inspect an HTTP request/response using browser dev tools; show headers and cookies.

5. Data & Content Layer — storage, processing, and formats
What it is: the actual information — databases, files, media, logs, user profiles.
Components / protocols: relational/noSQL databases, object storage (S3), CDNs (content distribution), file formats (JSON, XML, MP4). Backup systems and replication.
What happens: applications read/write data; CDNs cache media close to users for speed; databases index and replicate information for resilience.
Security risks: data breaches, unsecured backups, improper access control, weak encryption at rest.
Demo idea: show how a file uploaded to cloud storage is served by a CDN from a different region.

6. User Interface & Interaction Layer — the human face of cyberspace
What it is: browsers, mobile apps, command-line tools, chatbots — how users perceive services.
Components: UX/UI, authentication methods (password, OTP, biometrics), client-side scripts.
What happens: users interact, authenticate, authorize; client code renders content and sometimes executes logic locally (JS).
Security risks: phishing, credential theft, client-side XSS, malware on user devices.
Demo idea: create a mock phishing email to teach how to spot suspicious links and requests.

Step-by-step data flow example (putting it all together)

1. **User** opens browser (UI layer) and types https://bank.example.
2. **DNS** resolves the domain to an IP (network layer).
3. **Browser** opens a TLS session (transport + encryption).
4. **Request** goes over Wi-Fi, through ISP, across routers (physical + network layers) to the bank's load balancer (infrastructure).
5. **Load balancer** forwards request to an application server (service layer).

6. **App server** queries a database for account info (data layer).
7. **Database** responds; app composes HTML/JSON and returns it via HTTPS.
8. **Browser** renders page, user clicks "transfer" → client sends API call.
9. **Backend** checks authentication, writes transaction to DB, logs event, and sends notification — all coordinated across platform/cloud services.
10. **Monitoring systems** track success/failure and alert ops on anomalies (security & resilience).

In one line per layer

1. Physical — hardware that moves bits.
2. Network — IP addressing & routing (how bits find each other).
3. Transport — reliable/fast delivery between endpoints (TCP/UDP, TLS).
4. Application — services users interact with (HTTP, APIs).
5. Data — storage, databases, CDNs.
6. User Interface — how humans interact with services.

Communication and Web Technology in Cyberspace (Detailed Notes)

Cyberspace works only because **communication technologies** and **web technologies** connect devices, transfer data, and deliver online services. Both are the building blocks of how the internet functions.

1. What is Communication Technology in Cyberspace?

Communication technology refers to the **methods, devices, and systems** used to send data from one point to another over a network or the internet.

Why is it important?

Because without communication technology, users, servers, applications, and websites **cannot talk to each other**.

2. Components of Communication Technology

2.1 Transmission Media (How data travels)

Data travels from one device to another through:

a) Wired Media

- **Twisted pair cables** (used in LAN)
- **Coaxial cable**
- **Fiber optic cables**
  - Uses light, extremely fast
  - Backbone of the internet

b) Wireless Media

- **Radio waves** (Wi-Fi, Bluetooth)
- **Microwave transmission**
- **Satellite communication**
- **Mobile Networks (2G–5G)**

Example:

Your mobile sends a message → cellphone tower → internet backbone → server → delivered to another user.

2.2 Networking Devices

These devices help connect computers inside cyberspace:

- **Router** – sends data between networks
- **Switch** – connects devices inside a network
- **Modem** – converts digital signals for ISP
- **Repeater** – boosts signal strength over distance
- **Firewall** – protects the network

Example:

Router at home connects your mobile/laptop to the ISP → ISP connects to the global internet.

2.3 Internet Protocols (Rules of communication)
These are "languages" that computers use to talk.

Protocol	Purpose
IP	Addressing (every device gets an IP address)
TCP	Ensures reliable data transfer
UDP	Fast but less reliable (online games, streaming)
HTTP/HTTPS	Browsing websites
FTP	File transfer
SMTP/POP/IMAP	Email communication

Example:
When you open a website:
Browser → sends HTTP request → Server → returns webpage.

2.4 Internet Infrastructure
Communication technology also includes:

- **ISP (Internet Service Providers)**
- **Internet exchange points (IXPs)**
- **Submarine cables**
- **Data centers**
- **CDNs (Content Delivery Networks)**

These ensure the **global movement of data**.

3. What is Web Technology in Cyberspace?
Web Technology includes **tools, technologies, and protocols** that allow the creation and functioning of websites and web applications.

Components of Web Technology
3.1 Web Browsers

- Chrome
- Firefox
- Edge
- Safari

They allow users to access the web.

3.2 Web Servers

- Host websites and data
- Examples: Apache, Nginx, IIS

How it works:
Browser → sends HTTP request → Server → sends website back → Browser displays it.

3.3 Web Pages & Web Development Technologies
Front-end Technologies (Visible part)

- **HTML** – Structure
- **CSS** – Styling
- **JavaScript** – Interactivity

Back-end Technologies (Server-side)

- **Node.js**, **Python**, **PHP**, **Java**, **.NET**
- **Databases:** MySQL, MongoDB

These create and manage dynamic websites like Facebook, Amazon, YouTube.

3.4 APIs (Application Programming Interfaces)
Allow apps and websites to exchange data.

Example:

- Google Maps API in food delivery apps
- Payment gateway APIs in e-commerce sites

3.5 Web Hosting & Cloud Technology

Websites run on cloud platforms like:

- AWS
- Google Cloud
- Microsoft Azure

They provide compute, storage, and security.

4. How Communication + Web Technology Work Together in Cyberspace

Step-by-Step Flow

1. **User connects through communication network**
   (Wi-Fi / Mobile Data / Fiber)
2. Data reaches **router → ISP → internet backbone**
3. **DNS** converts website name to server IP.
4. Browser sends **HTTP/HTTPS request** to server.
5. Server processes and sends back webpage.
6. Browser renders it using **HTML, CSS, JS**.
7. User interacts with site → more requests sent.
8. Cloud servers, APIs, databases support the service.
9. Communication networks continuously deliver data packets.

5. Real-Life Example: Opening www.amazon.in

1. Your mobile uses **Wi-Fi or 4G** (communication tech).
2. Router → ISP → internet (communication infra).
3. Browser sends **HTTPS request** (web tech).
4. DNS finds Amazon's IP address.
5. Amazon server responds with HTML/CSS/JS.
6. Browser loads the Amazon homepage.
7. API calls fetch product list.
8. Web technologies render the interface.
9. Communication tech delivers product images, prices, etc.

Conclusion

- **Communication Technology = How data travels**
- **Web Technology = How websites and applications work**
- **Both together create Cyberspace**, enabling billions of users to communicate, shop, learn, and interact online.

What is Internet?

The **Internet** is a **global network of millions of computers and devices** that are connected together to share information and communicate with each other.
It is also called a **network of networks**.

Simple definition:

Internet is a worldwide system that connects computers so people can send information, access websites, communicate, and use online services from anywhere in the world.

Key Features of the Internet

1. Global Connectivity

It connects people and devices across countries, continents, and oceans.

2. Uses Standard Protocols

All communication happens using common rules like:

- **TCP/IP**
- **HTTP/HTTPS**
- **DNS**

3. Decentralized Network

No single person or company owns the whole internet.
It is operated by:

- ISPs
- Governments
- Universities
- Private organizations

4. Provides Multiple Services

The internet enables:

- Websites
- Emails
- Cloud storage
- Social media
- Online shopping
- Video streaming
- Voice/video calls (VoIP)

Easy Example to Understand the Internet

Imagine your phone is connected to your Wi-Fi.
Your Wi-Fi connects to your ISP.
Your ISP connects to the global network.
This global network connects to servers of Google, YouTube, Facebook, banks, colleges, etc.
So when you type **www.google.com**:

1. Your device → router → ISP
2. ISP sends your request through cables/fiber
3. Servers of Google receive it
4. Google sends the webpage back
5. Your browser displays it

All this happens in **less than a second**.

In One Line:

The Internet is the global system that allows computers around the world to communicate and share information using standard technologies.

What is the World Wide Web (WWW)?

The **World Wide Web (WWW or Web)** is a **collection of websites and web pages** that you can access through the internet using a browser.

Simple definition:

The Web is a system of web pages linked together using hyperlinks and accessed through the internet.

It is one of the **major services** provided *on* the internet (just like email, chat, cloud storage).

Key Points About the WWW

1. It is made of Web Pages

Web pages contain:

- Text
- Images
- Videos
- Audio

- Links

These pages are written using **HTML**, styled with **CSS**, and made interactive with **JavaScript**.

2. Accessed Using Web Browsers

Examples:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

3. It Works on Client–Server Model

Client (browser) sends a request → **Web server** sends the webpage back.

Example:

1. You type **www.amazon.in**
2. Browser requests Amazon's server
3. Server sends homepage
4. Browser displays it

4. Uses HTTP/HTTPS Protocols

- **HTTP** → HyperText Transfer Protocol
- **HTTPS** → Secure version of HTTP

These protocols define how web pages move between client and server.

5. Uses URLs to Locate Websites

Every web page has an address called a **URL** (Uniform Resource Locator).

Example:

https://www.youtube.com/watch?v=12345

Difference Between Internet and WWW

Internet	**WWW**
Global network of computers	Collection of web pages on the internet
Hardware + infrastructure	Software/service running on internet
Includes email, cloud, VoIP, FTP, etc.	Only websites/web content
Bigger system	Part of the internet

Internet = roads
WWW = cars and shops on the roads

Example to Understand the Web

When you open Instagram.com:

- You are using the **internet** to send/receive data.
- You are accessing **Web pages** (part of the WWW).

The World Wide Web is a system of interlinked webpages accessible on the internet using a browser.
Internet Infrastructure for Data Transfer and Governance.

The Internet works because of a **huge global infrastructure** that enables data to flow from one device to another and a **governance system** that ensures smooth functioning, rules, and coordination across countries.

We can divide it into 2 parts:

1**Infrastructure for Data Transfer (How data moves)**
2 **Infrastructure for Internet Governance (Who manages & regulates)**

1. Infrastructure for Data Transfer
Data transfer means **how information (messages, videos, emails, web pages)** travels from one device to another across the world.
This requires multiple components:

1.1 End-User Devices
These are the devices where data originates or ends:
- Mobiles
- Laptops
- Desktops
- IoT devices
- Smart TVs

These devices send data to the nearest network.

1.2 Access Networks (First point of connection)
These networks connect users to the internet:
- **Wi-Fi networks**
- **Mobile networks (3G / 4G / 5G)**
- **Broadband / Fiber-to-Home**
- **DSL / Cable connections**

Example:
Your mobile sends a message → to mobile tower → to ISP network.

1.3 ISPs (Internet Service Providers)
ISPs connect users to the global internet.
Examples:
- Jio
- Airtel
- BSNL
- ACT Fiber

ISPs provide:
- IP addresses
- Bandwidth
- Routing of data

1.4 Routers, Switches, and Gateways
These network devices route data between networks.
Router- Decides the best path for data packets.
Switch- Connects devices within a network (LAN).
Gateway- Connects different types of networks.

1.5 Data Centers & Web Servers
Data centers store:
- Websites
- Cloud services
- Emails
- Applications
- Databases

When you open a website, your request travels from ISP → routing devices → data center → server.

1.6 Submarine Fiber-Optic Cables
These are **massive undersea cables** connecting continents.
95% of international data transfer happens through these cables—not satellites.

They carry internet signals between countries at light speed.
Example routes:

- India to Europe
- India to Singapore
- India to the USA

1.7 Internet Exchange Points (IXPs)
IXPs are physical locations where ISPs connect directly to exchange data.
Example in India:

- NIXI (National Internet Exchange of India)

IXPs make:

- Faster data flow
- Lower cost
- Lower latency

1.8 CDNs (Content Delivery Networks)
CDNs store copies of popular content in multiple locations.
Examples:

- Cloudflare CDN
- Akamai CDN
- Google CDN

When you watch YouTube, the video is often delivered from a server **near your city**, not from the USA.

1.9 DNS (Domain Name System)
DNS converts website names (amazon.in) into IP addresses (52.98.00.1).
It works like the **phonebook of the internet**.
Without DNS, the web cannot function.

2. Internet Governance
Internet governance means **rules, organizations, and policies** that ensure the stable and secure functioning of the internet.
Important point:
No single country or company controls the entire internet.
Instead, different organizations coordinate standards.

2.1 ICANN (Internet Corporation for Assigned Names and Numbers)
ICANN manages:

- Domain names
- IP address allocation
- Top-level domains (.com, .in, .org)

2.2 IANA (Internet Assigned Numbers Authority)
Maintains key internet protocols:

- IP address blocks
- DNS root zone
- Protocol parameters

2.3 IETF (Internet Engineering Task Force)
Creates and maintains **technical standards**:

- TCP/IP
- HTTP
- SMTP
- DNS protocols

2.4 W3C (World Wide Web Consortium)
Created by Tim Berners-Lee.
It maintains **web standards**:

- HTML
- CSS
- XML
- Accessibility rules

2.5 National Governments
Each country has its laws for:

- Cyber security
- Data privacy
- Cyber Crime
- Digital payments
- Content regulation

In India:

- IT Act 2000
- CERT-In guidelines
- Data Protection Act 2023

2.6 ISPs and Telecom Companies
They follow government regulations for:

- Blocking harmful content
- Maintaining network security
- Traffic monitoring
- Complying with IT Act

2.7 International Justice Bodies
Used for:

- Cybercrime investigations
- Cross-border data requests
- Digital jurisdiction

Examples:

- Interpol Cybercrime Division
- Budapest Convention (cybercrime treaty)

Bringing It Together (Simple Explanation)
When a user opens a website:

1. Device → Access Network (Wi-Fi/4G)
2. Access Network → ISP
3. ISP → Routers → IXPs → Submarine cables
4. Data travels globally
5. DNS finds the server
6. Data Center → Web Server sends webpage
7. Content reaches the user
8. All this is guided by global internet governance organizations

Internet infrastructure enables data to move across the world, while internet governance ensures it happens securely, smoothly, and according to global standards.

Internet Society (ISOC)
The **Internet Society (ISOC)** is a **global, non-profit organization** founded in **1992** to ensure the **open development, evolution, and use of the Internet** for the benefit of all people across the world.

It supports the idea that the Internet should be:

- **Open**
- **Accessible**
- **Secure**
- **Reliable**
- **For everyone**

Main Purpose of Internet Society

ISOC works to make sure the internet remains:

- Free from monopoly or single ownership
- Technically robust
- Affordable and accessible
- Safe and secure for users

What Does ISOC Do

1. Supports Internet Standards

ISOC is the organizational home for:

- **IETF (Internet Engineering Task Force)**
- **IAB (Internet Architecture Board)**

These groups create technical standards like:

- TCP/IP
- HTTP
- DNS
- Email protocols

So ISOC indirectly ensures that **different devices and networks can communicate smoothly.**

2. Promotes Internet Access for All

ISOC works to:

- Expand internet access in developing countries
- Promote affordable and high-quality internet
- Bridge the digital divide

Example:
Building community networks in rural areas.

3. Works for Internet Security

ISOC encourages:

- Strong cyber security practices
- Protection of user data
- Secure routing (MANRS Initiative)
- Secure internet infrastructure

4. Advocates for Policy and Governance

ISOC participates in national and international discussions on:

- Digital rights
- Data privacy
- Cyber laws
- Net neutrality
- Online freedom
- Internet governance

It ensures the internet remains open—not controlled by any one government or corporation.

5. Education & Awareness

ISOC conducts:

- Workshops

- Training programs
- Cyber security awareness campaigns
- Community development programs

Purpose: to make Internet users informed and safe.

Global Presence

- ISOC has **100+ chapters** in different countries.
- India also has an **Internet Society India Chapter**.

These chapters work locally to improve internet access and digital literacy.

Why is ISOC Important?

ISOC ensures:

- Internet works smoothly worldwide
- No single company or country controls the internet
- Cyber security standards are maintained
- Web technologies stay open and interoperable
- Internet remains affordable and available to everyone

Without organizations like ISOC, the internet could become:

- Fragmented
- Unsecured
- Controlled by a few companies
- Expensive or restricted

The Internet Society (ISOC) is a global organization that works to ensure the Internet remains open, secure, and accessible to everyone.

Difference Between ISOC, ICANN, and IETF

These three organizations play important but different roles in managing and guiding the Internet.
They **do not control** the internet but ensure it runs smoothly, securely, and globally.

ISOC (Internet Society)

A global non-profit organization that promotes the open, secure, and accessible development of the Internet worldwide.

ICANN (Internet Corporation for Assigned Names and Numbers)

Manages domain names, IP addresses, and the DNS system (the address book of the internet).

IETF (Internet Engineering Task Force)

Develops and maintains the technical standards and protocols that run the Internet (e.g., TCP/IP, HTTP).

Feature / Role	**ISOC**	**ICANN**	**IETF**
Full Form	Internet Society	Internet Corporation for Assigned Names and Numbers	Internet Engineering Task Force
Type	Non-profit organization	Non-profit corporation	Open international community
Main Function	Promotes global internet development, access, and security	Manages domain names, IP addresses, DNS	Creates and maintains internet technical standards
Focus Area	Policy, advocacy, education, security	DNS management, domain system	Protocols and standards
Technical Work?	No direct technical work	Limited technical work (DNS)	Yes—creates standards like TCP/IP

Feature / Role	ISOC	ICANN	IETF
Controls Internet?	No	No	No
Key Output	Policies, awareness, governance	Domain names, IP allocation	RFCs (Request for Comments – technical standards)
Examples of Work	Net neutrality, internet accessibility	.com, .org, .in domain management	TCP/IP, HTTP, SMTP, DNS protocols
Governance Role	Broad internet governance	DNS governance only	Technical governance
Who Participates?	Members, chapters worldwide	Governments, ISPs, registrars	Engineers, researchers, tech experts

1. ISOC → Internet Policy + Advocacy
Helps keep the Internet open and accessible.
Supports organizations like IETF.
2. ICANN → Internet Address Manager
Allocates:

- Domain names (e.g., google.com)
- IP addresses
- Root servers

Responsible for the *DNS phonebook of the internet*.
3. IETF → Internet Protocol Maker
Creates and updates the rules that devices use to communicate:

- TCP/IP
- HTTP
- SMTP
- DNS standards

Everything technical runs on IETF standards.
One-line Difference

- **ISOC promotes and protects the open internet.**
- **ICANN manages domain names and IP addresses.**
- **IETF develops technical standards and protocols.**

Regulation of Cyberspace
Cyberspace is a virtual world where digital communication, online activities, and internet-based interactions happen.
Because billions of users, organizations, and governments operate online, **rules and regulations are needed** to ensure **safety, security, fairness, and lawful use** of the internet.
Regulation of Cyberspace simply means:
The laws, policies, standards, and organizations that govern behavior and activities on the internet.
It controls what is allowed, what is prohibited, and who is responsible in the digital world.

Why Do We Need Regulation in Cyberspace?
Cyberspace has no physical boundaries. Crimes can be done from any country.
Regulation is needed to:

- Prevent cyber crimes
- Protect user data
- Stop misuse of social media
- Maintain national security
- Control harmful content
- Protect children online

- Ensure ethical and responsible use of the internet

Components of Cyberspace Regulation
Cyberspace is regulated at 3 major levels:

1. National (Country-Level) Regulations
Each country creates cyber laws to govern online activities.
Examples:
In India:
- **IT Act 2000** – main cyber law
- **Data Protection Act 2023** – user data protection
- **Cyber Crime Sections under IPC**
- **CERT-In Directives** – for cyber security incidents
- Rules for:
    - Social media
    - Digital payments
    - E-commerce
    - Online content moderation

Purpose:
- Reduce cybercrime
- Define punishable offenses
- Protect citizens and businesses

2. International Regulations
Since cyberspace is global, international cooperation is essential.
Key international bodies:
- **ITU (International Telecommunication Union)**
- **Interpol Cybercrime Division**
- **UN (United Nations)**
- **Budapest Convention on Cybercrime** (for cross-border cybercrime cooperation)

These bodies promote:
- Common cyber security standards
- Information sharing
- Global cybercrime investigation support

3. Technical and Industry-Level Regulations
These are standards created by technical organizations to ensure secure functioning of the internet.
Major organizations:
- **ICANN** – domain names & IP addresses
- **IETF** – internet protocols & standards (TCP/IP, DNS, HTTP)
- **W3C** – web standards (HTML, CSS)
- **ISO** – security standards (ISO 27001)
- **ISOC** – internet openness & policy

These bodies ensure:
- Interoperability
- Secure communication
- Reliability of networks
- Uniform global standards

Areas Covered Under Cyberspace Regulation
Regulation typically includes:
1. Cyber Crimes
- Hacking
- Identity theft

- Phishing
- Ransomware
- Fraud
- Cyberstalking
- Voyeurism
- Child pornography
- Online harassment

2. Data Protection and Privacy
Rules for:
- Collection of personal data
- Storage
- Sharing
- Consent
- User privacy rights

Example:
India's **Digital Personal Data Protection Act (DPDP-2023)**.

3. Content Regulation
Control over:
- Fake news
- Hate speech
- Terrorist content
- Deepfakes
- Violent/obscene material

Platforms like Facebook, YouTube, and Instagram must follow guidelines.

4. Cyber Security Requirements
Mandatory security controls for:
- Banks
- Government websites
- Critical infrastructures
- Online payment systems

Includes:
- Encryption
- Secure authentication
- Network monitoring

5. E-Commerce and Digital Transactions
Regulation ensures:
- Safe online shopping
- Consumer protection
- Secure digital payments
- Prevention of financial fraud

6. Social Media Governance
Rules for:
- User verification
- Removing harmful content
- Reporting mechanisms
- Fake account identification

Challenges in Regulating Cyberspace
- **Borderless nature** of the internet

- **Different laws** in each country
- Difficulty in tracking cyber criminals
- Privacy vs. surveillance debates
- Fake accounts and anonymous users
- Rapidly evolving technologies (AI, IoT, Metaverse)

Regulation of cyberspace refers to the legal, technical, and policy frameworks that ensure safe, ethical, and lawful use of the internet.

What is Cyber Security?
Cyber Security is the practice of **protecting computers, networks, data, and digital systems** from unauthorized access, attacks, damage, or misuse.

Cyber security means protecting digital devices and information from cyber threats.

Why Cyber Security is Important?
Because today almost everything is connected to the internet—banking, education, government services, shopping, communication, health records.
If these systems are attacked, it can cause:
- Data theft
- Financial loss
- Privacy violations
- Service shutdown (e.g., banking systems going down)
- Harassment and social media misuse

Goals of Cyber Security – CIA Triad
The three main objectives are:
1. Confidentiality
Protecting information from unauthorized access.
Example: passwords, encryption.
2. Integrity
Ensuring data is not modified or altered.
Example: digital signatures, hashing.
3. Availability
Ensuring systems and information are available when needed.
Example: backups, uptime, disaster recovery.

Common Cyber Threats
1. **Malware** – viruses, worms, trojans
2. **Phishing** – fake emails/messages to steal information
3. **Ransomware** – locking data for ransom
4. **Social Engineering** – tricking people to reveal passwords
5. **Password attacks** – brute force, credential stuffing
6. **Insider threats** – employees misusing access
7. **DDoS attacks** – overwhelming a website to shut it down

Cyber Security Measures (How We Protect)
Technical Measures
- Firewalls
- Antivirus & Anti-malware
- Encryption
- Multi-factor authentication
- Network monitoring
- Security patches & updates

Administrative Measures

- Security policies
- Access control
- Regular audits
- Employee training

Physical Measures

- CCTV, locks, restricted access, biometric entry

Domains of Cyber Security

1. **Network Security**
2. **Application Security**
3. **Cloud Security**
4. **Data Security**
5. **Endpoint Security** (laptops, mobiles)
6. **Identity & Access Management**
7. **Operational Security**
8. **Incident Response and Forensics**

Cyber Security in Daily Life

- Using strong passwords
- Avoiding unknown links
- Updating apps regularly
- Using secure Wi-Fi
- Enabling screen locks
- Being careful on social media

If you lock your house to keep thieves out, cyber security is like locking your computer, mobile, and data to protect it from online thieves.

Here are the **8 major domains of Cyber Security** you should teach:

1. Network Security

Protects **networks and data during transfer** from unauthorized access or attacks.

This involves securing computer networks from intruders like hackers and malware. It uses tools such as intrusion detection systems, firewalls, and VPNs. These tools protect the network's perimeter. For example, a firewall blocks unauthorized access while allowing outward communication.

Includes:

- Firewalls
- Intrusion Detection Systems/Intrusion Prevention Systems (IDS/IPS)
- Virtual Private Networks (VPN)
- Secure network design
- Wi-Fi security (WPA2/WPA3)

Purpose:

To prevent hackers from entering or interrupting network communication.

2. Information / Data Security

Protects **data in all forms**—digital or physical.

Also known as data security, this type is dedicated to protecting data's confidentiality, integrity, and availability. It uses techniques such as encryption, access control, and data loss prevention to safeguard data stored and in transit. For instance, if you encrypt sensitive data files on a corporate network, it will prevent data leakage in case of a breach.

Includes:

- Encryption
- Data masking

- Access control
- Data classification
- Data loss prevention (DLP)

Purpose:
Ensures confidentiality, integrity, and availability of information.

3. Endpoint Security
Secures individual devices such as:
- Computers
- Laptops
- Mobile phones
- Tablets
- IoT devices

Includes:
- Antivirus
- Mobile device management (MDM)
- Patch management
- Application permissions

Protecting devices such as computers, smartphones, and tablets from cyber threats falls under endpoint security. It includes antivirus software, intrusion prevention systems, and comprehensive management of endpoint devices to prevent unauthorized access and data breaches. For instance, managing updates on all devices so that they are protected against newly discovered vulnerabilities.

Purpose:
Protects devices that act as entry points to the network.

4. Application Security
Secures software applications from vulnerabilities.
This domain focuses on protecting software and devices from threats. It involves employing secure coding practices, performing regular vulnerability assessments, and utilizing web application firewalls to defend applications against cyber exploits. A common application security measure is the regular updating and patching of software.

Includes:
- Secure coding
- Testing (static/dynamic analysis)
- Authentication & authorization
- Input validation
- Patch updates

Purpose:
Prevent attacks like:
- SQL Injection
- Cross-Site Scripting (XSS)
- Buffer overflow

5. Cloud Security
Secures data, applications, and infrastructure in cloud environments.
This security type protects data, applications, and infrastructures involved in cloud computing. It often involves a shared responsibility between cloud service providers and clients to secure data hosted in the cloud. For instance, providers might use encrypted data storage solutions.

Includes:
- IAM (Identity & Access Management)
- Data encryption in cloud
- Virtual firewalls
- Compliance policies

- Multi-tenant security

Purpose:
Protects cloud-based storage, virtual machines, and online services (AWS, Azure, Google Cloud).

6. Identity and Access Management (IAM)
Manages **who can access what** in an organization.
Includes:

- User authentication (password, biometrics)
- Role-Based Access Control (RBAC)
- Multi-factor authentication (MFA)
- Privileged Access Management (PAM)

Purpose:
Ensures only authorized users can access sensitive resources.

7. Operational Security (OpSec)
Focuses on processes, policies, and procedures for secure operations.

Operational security handles the processes and decisions for managing and protecting data assets. It covers managing permissions and access. The aim is to ensure that confidential information is only granted to authorized users. It involves options such as using multi-factor authentication to access corporate databases.
Includes:

- Security policies
- Regular audits
- Incident response plans
- Change management
- Risk assessment

Purpose:
Ensures daily operations remain secure and consistent.

8. Disaster Recovery & Business Continuity
Ensures systems continue working during and after a crisis.
Includes:

- Backup and restore
- Redundancy
- Failover systems
- Disaster recovery plans
- Continuity planning

Purpose:
Minimizes downtime due to:

- Cyber-attacks
- Hardware failure
- Natural disasters

Optional but Modern Domains

9. IoT Security
Protecting smart devices like CCTV, smart home devices, industrial sensors.
IoT security focuses on safeguarding interconnected devices and networks within the IoT ecosystem. This includes securing devices such as smart home gadgets, industrial machines, and other connected devices from cyber threats. For example, you can secure a smart home system with strong authentication methods.

10. Cryptography

Securing data using encryption, hashing, digital signatures.

11. Forensics & Incident Response
Investigating cybercrimes and responding to attacks.

1. Network security
Network security safeguards communication infrastructure, including devices, hardware, software, and communication protocols. It protects **data integrity**, confidentiality, and availability as information travels over a network and between network-accessible assets, such as a computer and an application server.
Network security also encompasses a broad collection of technologies, policies, people, and procedures. These focus primarily on preventing known threats from infiltrating the communication infrastructure.
For example, **firewalls** filter incoming and outgoing traffic, acting as a first line of defense by identifying familiar attack types, suspicious activity, or unauthorized access attempts based on pre-defined rules. The idea is that firewalls already know what to expect and have the capability to block these threats before they can cause harm.
However, network security tools must also include an element of detection. Firewalls and other network security solutions must be able to identify unfamiliar or new threats and, through integration with other systems, respond appropriately to mitigate the risk. To strengthen this posture, organizations also need continuous visibility into emerging **network security threats and vulnerabilities,** ensuring protections evolve alongside new attack techniques.

2. Information security
Information security, or InfoSec, is the practice of protecting information. It refers to the tools and processes for preventing, detecting, and remediating threats to sensitive information, whether digitized or not.
InfoSec is closely related to data security — a subset that specifically protects digitized data stored in systems and databases or transmitted across networks. Both disciplines share three primary objectives:

1. **Confidentiality:** Ensuring confidential information remains a secret.
2. **Integrity:** Protecting information from being altered, manipulated, or deleted.
3. **Availability:** Making information readily accessible to those who need it.

Therefore, information and data security solutions safeguard against unauthorized access, modification, and disruption. A key aspect of both disciplines is the need to scrutinize information, allowing organizations to classify it by criticality and adjust policies accordingly.
For example, data loss prevention (DLP) tools automatically discover and classify data as it's created. They also monitor, detect, and prevent unauthorized data sharing or extraction, ensuring valuable information remains secure within the organization.

3. Cloud security
Cloud security refers to the technologies, policies, and procedures that protect data, applications, and services hosted in private and public cloud environments. It ensures sensitive information is safe from **data breaches** and other vulnerabilities, whether stored in public, private, or hybrid clouds. Cloud security solutions are often versions of on-premises solutions that are specifically for the cloud. As such, cloud security can be a seamless extension of an organization's network security.
One of cloud computing's biggest security challenges is providing users with safe, frictionless access to their most essential applications. Cloud-based services are available off-premises, but the devices used to reach them are typically unprotected.
Organizations often mitigate security risks using **identity and access management** (IAM), a key strategy that ensures only authorized users can access specific resources. IAM solutions are not limited to cloud environments; they are integral to network security as well. These technologies include robust authentication methods, **multi-factor authentication (MFA)**, and other access controls, all of which help protect sensitive data and systems across both on-premises and cloud-based infrastructures.

4. Endpoint security

Endpoint security focuses on protecting the devices that serve as access points to an organization's network, such as laptops, desktops, smartphones, and tablets. These devices, or endpoints, expand the attack surface, providing potential entry points for cybercriminals to exploit vulnerabilities and infiltrate the broader infrastructure.

To reduce risk, organizations must apply the right security solutions to each endpoint, ensuring protection is tailored to the specific device and its role in the network. For example, laptops used by remote workers may require antivirus software and multi-factor authentication to **prevent malware attacks** or unauthorized access.

A related subset of endpoint security is **mobile security**, which specifically addresses the vulnerabilities of mobile devices. As employees increasingly use smartphones and tablets for work, securing these endpoints becomes critical to protecting the entire network. Security solutions, such as mobile device management, help organizations manage and secure these devices, preventing them from becoming weak links in the cybersecurity chain.

5. Application security

Application security refers to the technologies, policies, and procedures at the application level that prevent cybercriminals from exploiting application vulnerabilities. It involves a combination of mitigation strategies during application development and after deployment.

For instance, a **web application firewall (WAF)** monitors and filters traffic between applications and the outside world, blocking malicious activity like code injections or **cross-site scripting attacks**. With robust application security, organizations can ensure their software remains protected against threats that target the app and the sensitive data it processes and stores.

6. Zero trust security

Zero trust is a modern cybersecurity model that assumes no user or system, whether inside or outside the network, is automatically trustworthy by default. Instead, organizations continuously verify access to data and resources through strict authentication protocols.

Unlike traditional security models, which take a "castle-and-moat" approach, zero trust monitors more than just the perimeter. It enforces granular security controls across all endpoints, applications, and users, preventing unauthorized lateral movement. In other words, users can't freely roam inside the network without reconfirming their identity whenever they request access to a particular resource.

7. Operational technology (OT) security

OT security, which uses the same solutions and techniques as IT environments, protects the safety and reliability of system technologies that control physical processes in a wide range of industries. This includes critical infrastructure like manufacturing systems, energy grids, and transportation networks, where a security breach could result in significant damage, but has come to include banking systems and others as well.

Traditionally, security in these environments wasn't necessary. Most operational technologies weren't connected to the outside world, so they didn't require protection. Now, as IT and OT converge, they're increasingly exposed to malicious activity.

IoT security also focuses on protecting connected devices — but on a broader scale. IoT devices range from sensors in industrial equipment to smart thermostats in homes. Because they're web-enabled, these access points expand the attack surface. Plus, since they often have limited security capabilities, they're vulnerable entryways for cybercriminals to exploit.

To address this, IoT security solutions focus on device authentication, **encryption**, and **network segmentation**, ensuring secure communication and preventing unauthorized access. Organizations must monitor these devices closely and implement strong access controls to minimize risks.

The inclusion of IoT security into the traditional industrial world of OT has introduced a new concept: **cyber-physical systems and their security.**

What is a Cyber Threat?

A **cyber threat** is any **potential danger** or **possible attack** that can harm a computer system, network, or data.

A cyber threat is the possibility that someone may use digital techniques to damage systems, steal data, or disrupt online services.

Key Points to Explain
1. Cyber Threat = Potential Harm
It does not mean the attack has already happened.
It means **there is a risk** that an attack *could* happen.

2. Cyber Threats Target
- Computers
- Mobile phones
- Networks
- Websites
- Cloud systems
- Data

3. Who Creates Cyber Threats?
- Hackers
- Cyber criminals
- Malware authors
- Insider employees
- Terrorist groups
- Nation-states (cyber warfare)

Common Types of Cyber Threats
1. Malware
Software designed to harm a device.
Examples: virus, worm, trojan.
2. Phishing
Fake emails/messages to steal passwords.
3. Ransomware
Locks your data and demands money to unlock it.
4. Denial-of-Service (DoS/DDoS) Attacks
Overloads a website so it stops working.
5. Social Engineering
Tricking people to give confidential information.
6. Insider Threat
Employees misusing access.
7. Zero-Day Threat
A new vulnerability that has not been fixed yet.
8. MITM (Man-in-the-Middle) Attack
Attacker secretly intercepts communication.

If you receive an email saying **"Click here to update your bank account"—**
That is a **cyber threat**, because clicking it may steal your password.

A cyber threat is a risk that someone may attack a digital system, steal data, or cause damage using cyber techniques.

What is a Cyber Attack?

A **cyber attack** is a **deliberate attempt** by an attacker (hacker) to damage, steal, disable, or disrupt a computer system, network, or data.

A cyber attack is an intentional action taken by a hacker to break into a system and cause harm.

Key Characteristics of a Cyber Attack

1. It is intentional

The attacker has a purpose—steal money, access data, or damage systems.

2. It causes harm

A cyber attack results in:

- Data theft
- System damage
- Service disruption
- Financial loss
- Privacy loss

3. It uses digital tools

Attackers use:

- Malware
- Hacking tools
- Scripts
- Social engineering

Cyber Attack vs Cyber Threat

Cyber Threat	Cyber Attack
Possibility of harm	Actual harm has occurred
Risk	Action
Example: Fake email seems suspicious	Example: User clicks it and loses password

Common Types of Cyber Attacks

1. Malware Attack

Virus, worm, Trojans infect a system.

2. Phishing Attack

Fake emails/messages stealing passwords.

3. Ransomware Attack

Data locked; attacker demands money.

4. Denial-of-Service (DoS/DDoS) Attack

Flooding a website until it crashes.

5. Man-in-the-Middle (MITM) Attack

Attacker secretly intercepts communication.

6. SQL Injection

Inserting malicious code into a database query.

7. Brute Force Attack

Trying all possible passwords until one works.

8. Zero-Day Attack

Exploiting a new vulnerability before it is fixed.

9. Spoofing Attack

Fake identity — email spoofing, IP spoofing.

Simple Example

If a hacker steals your Facebook password and logs into your account —
That is a cyber attack.
If someone sends you a fake login link —
That is a cyber threat.

A cyber attack is an intentional action by a hacker to break into systems and cause damage, steal data, or disrupt services.

Types of Cyber Threats (Risk Stage)

- Phishing emails
- Zero-day vulnerability
- Suspicious links
- Malware present on the internet
- Weak passwords
- Insider risk
- Open ports
- Social engineering attempts
- Unpatched software

These **do not cause harm yet**, but they *could*.

Types of Cyber Attacks (Action Stage)

- Ransomware attack
- DDoS attack
- SQL injection
- Account takeover
- Data breach
- MITM attack
- Brute force attack
- Malware infection
- Website defacement

These **cause real harm**.

Situation	Category
A thief is outside your house, checking locks	**Threat**
The thief breaks the lock and enters your house	**Attack**

Same in cyber security.

Cyber Threat = possibility of attack
Cyber Attack = actual execution of threat

A. PURE CYBER THREATS (Possibilities)

These are *risks* that may lead to attacks:

1. Zero-Day Vulnerability
A weakness in software not yet fixed.
2. Insider Threat
Risk of employees misusing access.
3. Weak Passwords
Risk of brute force attack.
4. Suspicious Phishing Emails
Risk of falling into scams.
5. Exposed Ports / Open Networks
Risk of unauthorized entry.
6. Misconfigured Cloud Systems
Risk of data exposure.

B. CYBER THREATS THAT LEAD DIRECTLY TO ATTACKS

These can be threats if they have not caused harm **yet**.

1. Malware (before installation)
Virus available on the internet = threat.

2. Ransomware File (before activation)
Downloader present on system but not executed.
3. MITM attempt (attacker trying to listen)
Preparation stage = threat.
4. DoS attempt (no damage yet)
Traffic build-up = threat.
⚠ **These become attacks only when executed.**

C. CYBER ATTACKS (Actual Execution)
These cause harm:
1. Malware Attack
Malware infects your device and damages files.
2. Ransomware Attack
Data gets locked; ransom demanded.
3. Phishing Attack
User enters password into a fake site → stolen.
4. DDoS Attack
Website becomes unavailable.
5. MITM Attack
Attacker intercepts your communication.
6. SQL Injection
Database hacked.
7. Brute Force Attack
Password cracked.
8. Data Breach
Sensitive information stolen.

1. Increasing Number of Cyber Attacks
Cyber attacks are growing every year—ransomware, phishing, data breaches.
Organizations struggle to keep up with new threats.
Challenge: Faster growth of attacks than defenses.

Phishing attacks have become more sophisticated. Attackers use personalized and contextually relevant bait to trick victims into providing sensitive information or downloading malware. These scams have evolved from generic emails to targeted attacks using information gathered from social media and other public sources.

2. Advanced and Evolving Threats
Attackers constantly upgrade their methods:

- Zero-day attacks
- AI-powered hacking
- Advanced persistent threats (APTs)

Challenge: Security systems become outdated quickly.

Cyber threats are constantly evolving as attackers develop new strategies and technologies. What worked as a defense yesterday may not be effective today. That makes it a constant race against criminals for cybersecurity professionals.

3. Lack of Cyber Awareness Among Users
Many people:

- Click unknown links
- Use weak passwords
- Share personal data easily

Challenge: Human error is the biggest cause of breaches.

4. Shortage of Skilled Cyber Security Professionals
Globally, there is a huge gap in:

- Ethical hackers
- Cyber analysts
- Incident response experts

Challenge: Not enough experts to protect systems.

5. Rapid Growth of IoT Devices
Smart devices (CCTV, Alexa, smart bulbs, sensors) are vulnerable.
Challenge: Hard to secure millions of connected devices.

The expanding network of IoT devices opens new vulnerabilities. Each connected device, from smart thermostats to industrial sensors, potentially provides a gateway for attackers. Securing these devices is a significant challenge due to their diverse nature and the difficulty of implementing uniform security standards across them.

6. Cloud Security Issues
Organizations shift to cloud services like AWS, Azure.
Problems:

- Misconfiguration
- Unauthorized access
- Data leakage

Challenge: Shared responsibility model is often misunderstood.

As organizations become increasingly dependent on cloud computing, protecting cloud-based resources becomes more complex. Providers and clients must strictly enforce security protocols to safeguard data stored and processed in cloud environments.

7. Data Privacy Concerns
Large amounts of personal data are collected (banks, hospitals, social media).
Challenge: Protecting privacy and complying with laws (GDPR, DPDP Act).

Navigating the myriad privacy and security regulations, such as GDPR, HIPAA, or PCI DSS, adds complexity to cybersecurity efforts. Compliance requires protecting sensitive information and adhering to specific procedural and technical standards, often varying by region or industry.

Website :- What is Cyber Security? Types, Importance & Threats
What is Cybersecurity? Different types of Cybersecurity | Fortinet
Discuss in detail the architecture of cyberspace and its role in communic..